

Direct-Answer Edit Success Is Not Enough: A Reasoning-Time Stress Test for Parametric Knowledge Editing

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Abstract

An edit can pass direct-answer validation yet lose control of the final answer when the model reasons. We evaluate each ROME edit on CounterFact twice—at zero thinking and after a natural chain—across six fixed checkpoints from two DeepSeek-R1-distill families. At the largest tested checkpoint in each family, edit success drops by 0.106 on Qwen-32B and 0.107 on Llama-70B. Among zero-thinking successes on Qwen-32B, old answers resurface in 19.3% of reasoned final answers and displace new answers outright in 9.2%, while the unedited base shows no detectable old-answer drift. Yet in the Qwen-32B reversions we examined, the edited association remains detectable at a cloze probe; separately, route analysis finds visible in-chain paths to old answers, motivating a chain-routing hypothesis. We next test for a chain-local control point by suppressing old-answer first tokens only during Qwen-32B reasoning, leaving answer logits untouched. The signed, dose-graded intervention offsets the edit-success loss and lowers the resurfacing rate from 19.3% to 8.5%; placebo and same-relation controls have near-null answer-level effects. Direct-answer edit success is not enough: edit evaluation needs a reasoning-time stress test, and the failure it reveals admits a chain-local causal control point.

1 Introduction

Parametric knowledge editing offers a targeted alternative to retraining: find the weights that carry one fact, change them, and leave the rest of the model alone (Meng et al. 2022). Whether the change took is then decided by asking the edited model the edited question and checking what comes back, sometimes alongside paraphrase and neighbourhood probes. That protocol carries an assumption it never tests—that an edit which answers correctly on demand will keep answering correctly wherever the model is actually used.

Deployed reasoning models do not answer on demand. They think first, in tokens, and the answer arrives after the chain. So certification happens at zero thinking while deployment happens after reasoning, and nothing in current practice measures the distance between the two. Does thinking undo editing?

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We test that question under one protocol. Each edit is applied on its own, evaluated at zero thinking and again after the model’s own native chain, then reverted before the next request, so requests cannot contaminate one another. Scoring reads the generated text with boundary-aware value matching rather than teacher-forced logits, and the same test runs at six fixed checkpoints drawn from two backbone lineages distilled under one reasoning recipe.

Our contributions are three.

A reasoning-time evaluation gap. At the largest tested checkpoint in each lineage, edit success is lower after a native chain than at zero thinking, and old answers resurface in reasoned final answers among edits that had already succeeded. An unedited base, a fixed canned thought, and temperature sampling each fail to reproduce the pattern, so it is neither background drift nor a decoding artifact.

A chain-local causal control point. Suppressing the old answer’s first tokens inside the think span—never touching the answer stage—raises edit success and lowers resurfacing against a placebo floor, a same-relation competitor, and no suppression at all, while the reverse-signed arm moves both endpoints the other way. We report this as a diagnosis of where the failure is steerable, not as a deployed defence.

Bounded transfer, honestly bounded. The intervention repeats at a second scale and under a second editor, and applies to the benchmark’s own paraphrase probes; its strict displacement endpoint does not separate from the untreated baseline, and we say so at the same prominence as the effects that do.

Between the failure and the intervention sits a diagnosis we deliberately leave open: in the reversions we examined the edited association is still detectable at a cloze probe, and native chains show visible paths back to the old value, but the test that would have identified natural-chain mediation failed its own carrier-validity gate and we withdraw that claim. What survives is narrower and, we think, more useful: direct-answer edit success is not enough, and the failure it hides admits a chain-local causal control point.

2 Related Work

How edits are evaluated. Editing methods are normally scored on reliability, generalization, and locality. He, Song, and Sun (2025) report that ten such methods degrade on multi-hop and sequential settings and propose Selective Con-

textual Reasoning, which retrieves and contextualizes an update instead of touching parameters. He et al. (2025) then benchmark editing under realistic autoregressive inference rather than teacher-forced decoding, across instruction-tuned and reasoning-oriented models, and find parameter-based editing performs poorly there. Hua et al. (2024) show with ReCoE that edits fail to propagate through the reasoning schemes that depend on them, and analyse the chains to explain why. Slocum et al. (2025) ask a related question from the other side—how deeply an implanted fact is believed—and find mechanistic editing implants shallowly. This prior evidence is cross-model and single-configuration: it establishes that editing looks worse under realistic decoding, not how a single edit behaves as its own thinking budget changes.

Editing under explicit reasoning. Huang et al. (2026) make the closest observation to ours in the multimodal setting, reporting near-perfect teacher-forced accuracy alongside collapsing grounded success once the chain is examined, and attributing it partly to chains that reject the injected fact. Mao et al. (2026) benchmark the adversarial direction, where maliciously edited knowledge corrupts downstream reasoning while general capability stays intact. Gema et al. (2025) document the more general hazard that longer reasoning can lower accuracy. Gekhman et al. (2026) supply the mechanism our diagnosis has to argue with: reasoning tokens aid recall both as a computational buffer independent of their content and through factual priming by related generated facts.

Intervening at decoding time. Bi et al. (2024) contrast edited against unedited next-token distributions to raise a model’s confidence in an edited fact, and Jung, Lee, and Lim (2026) suppress original-knowledge probabilities during decoding for lifelong editing. Both act on the distribution that writes the answer. Ours is scoped to the think span and leaves the answer stage untouched, which is what makes it a probe of where the failure is produced rather than a better decoder.

What this paper adds. Four properties are not addressed jointly by the work above: a budget-controlled paired contrast on one and the same edit; controls that separate edit-specific erosion from base drift, canned thought, and sampling; symmetric results at the largest fixed checkpoint of two backbone lineages; and a signed, dose-graded intervention confined to the think span that moves an answer it never touches.

3 The Reasoning-Time Evaluation Gap

3.1 Protocol and Metrics

We apply one ROME edit at a time to a CounterFact request (Meng et al. 2022), generate every configured thinking budget for that edit, and then restore the original weights before proceeding to the next request. This edit-generate-restore loop keeps requests from contaminating one another. Our headline comparison pairs zero thinking (B_0 , a pre-closed empty think span) with a native chain (B_3) under greedy decoding. We run the same protocol at six fixed checkpoints in the Qwen and Llama backbone lineages distilled under the DeepSeek-R1 recipe (DeepSeek-AI et al. 2025).

Scoring is generative rather than logit based. On answer and chain texts, we remove subject mentions before apply-

ing boundary-aware value matching over curated aliases. For value v , let $h_v(x)$ mark an alias hit in text x . Edit success is $ES_b = \Pr[h_{\text{new}}(a_b) \wedge \neg h_{\text{old}}(a_b)]$. Among requests successful at B_0 , permissive reversion RR_b records any old-value hit in the later answer, whereas strict lexical displacement RR_b^s additionally requires the new value to be absent. These are nested operational endpoints with different error profiles, not mathematical bounds on a latent semantic rate. Chain leakage CLR_b records an old-value mention in the chain without claiming semantic commitment. Locality is used only as a target-value non-leakage check on the neighbor answer, not as a general preservation measure.

We keep the two judge instruments distinct.¹

Conditioning reversion on zero-thinking success is treatment-independent for the later intervention study: a think-span processor is inactive when the think span is empty. In a separate replication audit, the treated-versus-no-suppression comparison covered 400 B_0 prompt/chain/answer tuples with 0 differences; the same-relation-control-versus-no-suppression comparison likewise covered 400 tuples with 0 differences. Each total comprises 200 cases and two probe types; arm-wise contrasts use case intersections.

3.2 The Gap

At the largest tested checkpoint in each family, the paired ES drops are .106 [.030, .182] on Qwen-32B and .107 [.032, .182] on Llama-70B (Table 1). We report all six fixed-checkpoint cells. The cellwise, unadjusted 95% paired case-bootstrap intervals exclude zero only at the largest tested checkpoint in each family. These cells delimit the headline scope—a scope statement, not a post-hoc selection—and we do not infer an ordered trend across checkpoints.

Among zero-thinking successes on Qwen-32B, permissive old-value resurfacing is 19.3% [12.6%, 26.9%], while strict lexical displacement is 9.2%. For the unedited base on the same requests, the paired old-answer drift over $n = 199$ cases is $-.0201 [-.0804, .0402]$, which does not support a matching base-model drift. This juxtaposition is consistent with an edit-specific pattern, but the two quantities differ in event definition and denominator; we make no formal contrast.

The similarity across the two tested backbone lineages disfavors a single-lineage idiosyncrasy, though not recipe-specificity: both share an R1 teacher and distillation recipe, and the comparison does not establish equality across lin-

¹The answer-commitment validation panel uses a neutral OLD/NEW/NEITHER final-answer task. Its strict rule has available-case Cohen’s $\kappa = .836$, agreement = .931, and precision/recall = .92/.85 against the three-judge majority; the permissive rule has precision/recall = .48/.96. The panel obtained votes for 87 of 120 stratified items; 33 were excluded by rate limits. Missingness is concentrated among rule-negative rows but does not make the available-case κ a lower bound; missingness sensitivity is deferred to the supplement. The separate route-census panel uses a neutral visible-route categorization task over 41 route instances from 33 unique facts and reports inter-judge Fleiss’ $\kappa = .813$; Section 4 defines its population. Both panels use one external judge-model family, creating shared-bias risk.

Backbone	Parameters	Marginal ES_{B_0}	Marginal ES_{B_3}	Paired ES drop [95% CI]	Permissive RR_{B_3} [95% CI]
Qwen	1.5B	.582	.607	−.026 [−.097, .051]	.061 [.018, .105]
Qwen	7B	.500	.530	−.030 [−.110, .050]	.080 [.030, .140]
Qwen	14B	.555	.515	.040 [−.030, .110]	.162 [.099, .234]
Qwen	32B	.595	.495	.106 [.030, .182]	.193 [.126, .269]
Llama	8B	.625	.610	.015 [−.060, .095]	.144 [.088, .208]
Llama	70B	.610	.503	.107 [.032, .182]	.175 [.105, .246]

Table 1: Reasoning-time evaluation across all six fixed checkpoints. ES levels are marginal; drops and intervals are within-case complete-pair estimates and need not equal marginal differences when rows are missing. For Qwen-32B, the marginal B_0 level uses 200 attempted rows and the paired change uses 198 complete pairs. The Llama-70B headline row uses the early-stop-filtered $n = 187$ analysis; because this filtering can be directional, we also report the unfiltered $n = 200$ sensitivity, whose paired drop is .110 and permissive RR is .167.

eages. Our claims concern the tested fixed checkpoints; separating reasoning post-training from backbone capability requires a matched parent–distill control, which remains future work.

3.3 Controls and Robustness

A dedicated Qwen-32B control gives permissive RR 0/.040/.208 under $B_0/B_{0P}/B_1$, respectively. The B_0 zero is structural under the success gate; the informative contrast is the separation between the canned-thought and native-chain arms. The B_{0P} arm is a fixed short canned thought, not a length-matched filler. It improves rather than erodes ES relative to B_0 , whereas the native B_1 chain reduces it.² This control distinguishes the empty think wrapper from the tested canned content, but it does not isolate length; Section 4 revisits the length-only account.

The gap also persists under temperature sampling across three fixed seeds.³

Binary unedited-base recall attenuates, but does not eliminate, the fixed-checkpoint size association. The regression is in the supplement. Finally, case-block fits provide supporting outcomes on the same source pool and six fixed checkpoints, not a second independent trend line. Both fits are in the supplement.

4 Causal Diagnosis

This section assembles the diagnosis that motivates the primary intervention in Section 5, separating observational evidence, a failed replay test, and boundary evidence.

4.1 Installation Sanity: The Edited Association Remains Cloze-Detectable

For the selected Qwen-32B reversion ($n = 23$), we reapplied ROME and compared the new and old objects’ first-token logits at the original cloze; the new token had the higher logit in 23/23 cases. This is only an installation sanity check: no control, no interval, close to tautological

²The paired ES drops are −.090 [−.150, −.030] from B_0 to B_{0P} and .130 [.060, .205] from B_0 to B_1 .

³Pairing B_0 and B_3 within case and seed gives a complete-case ES drop of .117 [.067, .169] over 199 of 200 cases and 597 of 600 seed pairs. These fixed seeds do not identify a sampling-seed population.

because ROME optimizes the target logit at the rewrite prompt, and conditional on the selected reversion sample. The supplement names the missing per-item artifact, results/probe/logitlens_cf200_ROME_B3.jsonl, whose case identifiers and first-token logits would be needed to audit sample overlap and membership composition and to reproduce the case-level tally; absent it, we place no quantitative weight on this observation downstream.

4.2 Visible In-Chain Routes

From a frozen pool of 101 candidate chains drawn from six checkpoints across both backbone lineages, with greedy, short-budget, and sampled generations represented, an answer-only membership gate retained 41 OLD-committed route instances representing 33 unique facts. The gate assigned the remaining candidates 43 NEW and 17 NEITHER. Within this pool, these rejections show that lexical candidacy is not equivalent to OLD membership by judge majority. Separately, Section 3.1 reports available-case precision .48 for the permissive matcher on its stratified validation sample; we do not transport that estimate to this pooled census. Within this outcome-conditioned failure surface, the route counts are 27 Bridge, 11 Recall, 3 Reflective-override, and 0 Associative. Route-vote agreement was high, with route-panel Fleiss’ $\kappa = .813$; Table 2 carries the per-instance vote splits. The answer-only membership gate is a distinct task.⁴ In a separate post-hoc A3 candidate crosswalk, subject-scrubbed CLR appears in 13/13 majority-OLD items and 18/20 non-OLD candidates; within that crosswalk, CLR is non-sufficient for OLD commitment. Because the census conditions on committed-OLD outcomes, it describes observed failures rather than a predictor or causal mediator; the baseline-CLR split below is a separate contrast. This defines the route population promised in Section 3.1.

4.3 The Chain-Routing Hypothesis and a Failed Replay Vehicle

The prevalence of Bridge and Recall routes motivates a chain-routing hypothesis consistent with the factual-priming mechanism of Gekhman et al. (2026), in which related facts can serve as semantic bridges during recall. We next describe

⁴Its Fleiss’ $\kappa = .958$. Both panels use one external judge-model family, so their errors should not be treated as independent.

Figure placeholder. Paired B_0 -to- B_3 ES changes at the six fixed checkpoints, followed by the Qwen-32B base, canned-thought, and sampling controls.

Figure 1: The reasoning-time stress test exposes a gap hidden by direct-answer evaluation. Table 1 provides the complete numeric record; Section 5 tests whether the failure admits a chain-local causal control point.

Table placeholder: corrected route counts and agreement.

Table 2: Corrected visible-route counts and agreement after answer-only membership gating.

the prospectively specified attempt to test whether fixed replay of a model’s own chain was stable enough to carry a mediation experiment.

The replay conjunction succeeded in 9/18 cases, below the required 15, so the vehicle-validity gate is permanently FAIL; the stop rule prevented the planned semantic-control arms from running, and we withdraw the planned natural-chain mediation claim. Both prerequisites limited the conjunction: fresh- B_0 strict direct-answer success held in 14/18 cases, while replayed answers received an OLD majority in 12/18.⁵ A non-rescuing post-hoc audit found that only 13/18 frozen source answers committed to OLD; separately, only 1/5 F2-sourced cases reproduced fresh- B_0 strict direct-answer success. These diagnostics constrain the proposed vehicle but do not identify a single mechanism; because the gate failed before the semantic-control arms ran, natural-chain mediation is neither confirmed nor refuted.

The control-point claim in Section 5 does not rely on the replay vehicle: it intervenes on live chains, does not directly intervene on answer logits, and its controls do not route through replay. The complete replay-gate table is in the supplement.

4.4 The Length-Only Account

The B_{0P} control in Section 3.3 is a short task-independent canned thought, not a length-matched filler. At B_3 , old-answer first-token suppression in the think span yields marginal ES .631, compared with the Qwen-32B B_0 marginal level of .595.⁶ Because suppression may alter realized chain length, these levels do not form a length-matched contrast. Separately, permissive RR at B_1 is .208, descriptively similar to .193 at B_3 . Because these estimates come from separate runs with run-specific B_0 gates, we infer neither saturation nor an ordered trend. The short

⁵The replay answer judges had Fleiss’ $\kappa = .9195$; this is distinct from both membership and route agreement above.

⁶The corresponding marginal denominators are 198 and 200, respectively.

task-independent thought yields permissive RR .040, but an equal-length content-free filler remains untested. Accordingly, the content-independent computational-buffer account of Gekhman et al. (2026) remains a live alternative. Section 5 bears on this account without settling it: at the same nominal budget B_3 , think-span-only suppression of old-answer first tokens improves edit success and lowers permissive reversion against unsuppressed chains, while the strict displacement endpoint does not separate from that baseline. Because realized chain lengths under suppression were not audited, that contrast constrains a length-only reading rather than eliminating it.

4.5 CLR-Stratified Enrichment and a Dose-Limited Boundary Test

For T (old-answer first-token suppression) versus N (no suppression), the interval for the T-minus-N RR difference within N-arm-CLR strata excluded zero for CLR-positive but not CLR-zero cases.⁷ The CLR-zero stratum had an N-arm RR floor of 0; without an interaction test, this is within-sample enrichment and cannot establish that visible CLR is required or identify mediation.

An independent concept-direction experiment provides a dose-limited boundary test. It compares old-object direction suppression (S) with a cross-case shuffled-direction control (P_{shuf}). On the confirmatory arms, the pre-specified primary RR contrast was compatible with zero, even though the in-span CLR manipulation check fell; ES remained null-compatible. Against no steering, RR was null-compatible while ES was positive; a secondary concept-versus-shuffle contrast on the pre-specified LocAcc harm metric was negative. The complete endpoint set is in the supplement. Reducing measured old-object expression therefore did not yield detectable confirmatory RR repair in this dose-limited test, and it does not locate a distinct commitment layer.

In the selected Qwen-32B reversion sample, the edited association remains cloze-detectable; separately, the answer-gated census labels Bridge or Recall routes in many observed failures. Neither observation identifies natural-chain mediation. Section 5 tests whether think-span-only old-answer first-token suppression changes the answer without a direct

⁷The paired strata contained 106 CLR-positive and 90 CLR-zero cases; these are arm-paired stratum totals, not RR denominators. Among cases with RR defined under the B_0 gate in both arms, T-minus-N RR was $-.289 [-.444, -.156]$ versus $+.014 [.000, .041]$, with bootstrap $p = .0002$ and $p = .72$.

answer-stage intervention—the test behind our chain-local causal control point.

5 A Chain-Local Causal Control Point

5.1 A Signed, Dose-Graded Think-Span Intervention

During the think span only, we add a fixed logit penalty to the first tokens of the old value’s aliases; decoding is otherwise unchanged, and no penalty is applied while the answer is written. Five arms share one edit, one case set, and one decoding configuration, differing only in which tokens are penalized: no suppression (N), the old answer (T), the new answer (D), a frequency- and length-matched placebo word (P), and a same-relation strong competitor (C). The first four arms are the frozen protocol; the competitor arm was added afterwards, with its donor list recorded before that arm ran.

The primary contrast was fixed in advance as T versus P—the margin of the old-answer effect over the placebo floor—in a protocol recorded with its placebo donors before any suppressed arm ran. Paired edit success and permissive reversion are the primary endpoints; strict lexical displacement is secondary, and we report its unadjusted bootstrap p verbatim. Every arm-by-endpoint contrast appears in Table 3 without multiplicity adjustment, and we claim none.

Three properties keep this from restating the scoring rule. First, the penalty and the measurement occupy different spans: nothing is penalized in the answer text on which edit success, reversion, and displacement are scored. Second, a scope ablation shows what directly pushing the answer would look like—extending the same penalty to the answer span raises zero-thinking edit success from .580 to .727 over 100 cases, because the old value is then suppressed where the answer itself is written, whereas think-scoped suppression leaves that level untouched. Third, chain leakage is a manipulation check on whether the penalty acted inside the span, not an outcome.

Conditioning reversion on zero-thinking success is treatment-independent here for a structural reason: at B_0 the think span is empty, so a think-scoped processor has nothing to act on. This battery has no byte-identity audit of its own—the arm-wise B_0 audit in Section 3.1 belongs to the MEMIT replication—and arm-wise n differences here reflect harness errors only.

Clamp strength was varied over 0, 2, 4, 8, 12, and 16, with 8 used throughout the reported arms. Graded across these six settings, the in-span leakage check declines at every step, reaching .082 at the strongest clamp, while edit success and permissive reversion are not ordered across the same settings.⁸

The processor fires only on probes for the edited subject and relation, and it needs the old value’s alias strings to do so; the Ethical Statement returns to that trigger condition.

⁸The zero-clamp point is the no-suppression arm itself; the five suppressed points come from a smaller separate run, and none of the six carries an interval. We therefore read them as a dose ladder rather than a calibrated dose-response.

5.2 The Control Point Moves the Untouched Answer

On the contrast fixed in advance, suppressing the old answer beats the placebo floor: paired edit success rises by +.122 [.051,.193] and permissive reversion falls by -.095 [-.171,-.029]. Because that floor is itself compatible with zero (+.015 on edit success, $p = .709$), the comparison against no suppression is close to the same quantity: paired edit success rises by +.138 [.082,.194] and permissive reversion falls by -.102 [-.170,-.042]. The arm-wise levels quoted in the abstract are therefore not an artifact of which control the treated arm is measured against.

Arm-wise, permissive reversion among zero-thinking successes is 19.3% without suppression and 8.5% with it.⁹ The no-suppression arm is the same Qwen-32B run as the largest Qwen cell of Table 1, which is why its level matches the one reported there.

5.3 What the Placebo and Competitor Arms Bound

Neither control arm moves the answer. Against no suppression, the placebo changes edit success by +.015 ($p = .709$), permissive reversion by .000 ($p = 1.00$), and strict displacement by -.009 ($p = .828$); the same-relation competitor changes them by +.015 ($p = .688$), .000 ($p = 1.00$), and -.009 ($p = .828$). These are compatible with zero; we set no equivalence margin, so they are not evidence of no effect.

Inside the chain the two controls are not inert. The manipulation check falls by -.276 [-.337,-.214] under old-answer suppression, to an arm level of .263; it also falls by -.071 [-.121,-.020] under the competitor and by -.051 [-.102,.000] under the placebo ($p = .054$). Both controls therefore perturb what the chain says while leaving the answer where it was. That dissociation is what this section rests on, and it is also why inertness may be claimed only at the answer level.

How active the competitor is at decoding time was audited in the transfer runs rather than here: under MEMIT on the Qwen-14B checkpoint the treated arm changed 109 probe rows against no suppression while the competitor changed 18, and in the ROME-32B paraphrase run the counts were 215 and 24 of 400. The five-arm battery has no changed-row audit of its own, so the competitor is described as a strong but largely inactive control on the strength of those two runs.

The direction arm completes the sign evidence: suppressing the new answer instead moves edit success by -.217 [-.289,-.144] and permissive reversion by +.086 [.019,.152]. The four arms order as the frozen prediction stated, though that ordering is a signed ladder and not a calibrated dose. One boundary is structural rather than statistical: every arm here is an intervened condition, so nothing in this battery shows that the chain route is required in ordinary, uninterrupted generation.

⁹Paired changes are complete-case within-edit contrasts and need not equal differences of arm-wise rates. Permissive reversion moves from 19.3% to 8.5% arm-wise while the paired change is -.102 [-.170,-.042]; edit success moves from .495 to .631 arm-wise while the paired change is +.138, and .495 plus +.138 does not equal .631.

Arm	Penalized tokens	n	ES_{B_3}	RR_{B_3}	$RR^s_{B_3}$
N	no suppression	198	.495	.193	.092
T	old-answer first tokens	198	.631	.085	.042
D	new-answer first tokens	196	.281	.309	.236
P	a matched placebo word	199	.507	.210	.113
C	a same-relation competitor	200	.510	.208	.112

Contrast	Endpoint	Difference	Interval
T-P	ES	.122	[.051, .193]
T-P	RR	-.095	[-.171, -.029]
T-P	RR^s	-.057	[-.114, -.010]
T-N	ES	.138	[.082, .194]
T-N	RR	-.102	[-.170, -.042]
T-N	RR^s	-.051	[-.102, .000]
T-C	ES	.121	[.051, .192]
T-C	RR	-.094	[-.160, -.028]
T-C	RR^s	-.057	[-.113, -.009]
P-N	ES	.015	[-.046, .081]
P-N	RR	.000	[-.057, .057]
P-N	RR^s	-.009	[-.047, .028]
C-N	ES	.015	[-.046, .076]
C-N	RR	.000	[-.056, .056]
C-N	RR^s	-.009	[-.047, .028]
D-N	ES	-.217	[-.289, -.144]
D-N	RR	.086	[.019, .152]
D-N	RR^s	.095	[.038, .162]

Table 3: Five-arm think-span battery on Qwen-32B at B_3 . Upper panel: arm-wise levels. Lower panel: within-case paired differences with case-paired bootstrap intervals at the 95% level. ES is scored on every attempted row, whereas RR and RR^s are conditioned on zero-thinking success, so the two panels rest on different denominators. Strict displacement RR^s is secondary and its unadjusted p values appear in the text. The in-span leakage check is a manipulation check rather than an outcome and is reported in the text. Intervals are unadjusted for multiplicity.

5.4 Endpoint Hierarchy and Multiplicity

The secondary strict endpoint separates the treated arm from both controls, by $-.057$ ($p = .029$) against the placebo and $-.057$ ($p = .032$) against the competitor, but not from the untreated baseline, where it is $-.051$ $[-.102, .000]$ with $p = .073$. That interval reaches zero. The comparison rests on 118 paired cases that clear the zero-thinking gate in both arms, and arm-wise strict displacement runs at $.092$ of gated cases without suppression against $.042$ with it, so the endpoint is thinnest where the effect is smallest. All p values in this section are unadjusted and we claim no correction; the strict endpoint is reported at the same prominence as the primary ones precisely because it is the weakest.

5.5 How Far the Guard Transfers

The intervention was re-run under three changes of setting, each with its own control and ceiling. Under ROME on the Qwen-14B checkpoint, against no suppression, paired edit success rises by $+.135$ $[.055, .215]$ and strict displacement also separates, $-.040$ $[-.081, -.010]$ with $p = .035$ —a second scale, not a second editor, and strict separation where the Qwen-32B battery had none.

Under MEMIT on the Qwen-14B checkpoint, against the same-relation competitor, paired edit success rises by $+.100$ $[.040, .160]$ while strict displacement does not separate, $-.027$ $[-.080, .027]$ with $p = .400$; in that run the competitor’s

own strict contrast against no suppression is $.000$ $[.000, .000]$, a floor rather than an equivalence result. That replication’s fresh no-suppression arm showed no reasoning-time drop, so it transfers the intervention and not the gap.

Under ROME on the Qwen-32B checkpoint, applying the same think-span suppressor to the benchmark’s own paraphrase probes raises strict lexical paraphrase success by $+.055$ $[.020, .093]$ over the competitor. This is active application to a paraphrase query, not portability to unguarded queries: a blind audit found 30 of 40 paraphrases valid, 11 of 20 cases valid on both, $\kappa = .444$; the secondary old-value comparison stayed directional at 110 gated cases against a frozen floor of 140. Its no-suppression levels resemble the battery’s but come from an independent run and do not check it.

We therefore position this as a mechanistic probe and an edit-aware decoding guard, not as a deployment-ready fix. Target-value non-leakage and general-benchmark cost are boundaries rather than transfer; Section 6 treats them and the question of whether guarding reasoning is desirable at all.

6 Discussion & Limitations

Our claims are statements about six fixed checkpoints rather than about scale in general: we report every cell and infer no ordered trend across them. Both lineages also share one teacher and one distillation recipe, so their agreement

bounds backbone idiosyncrasy but not recipe-specificity, as Section 3.2 states.

The headline rests on one editor and one dataset. MEMIT enters only through the transfer replication, and a second benchmark, a second edit family, and non-counterfactual edits are untested.

Our endpoints are lexical. Edit success, reversion, and displacement read generated text with boundary-aware matching over curated aliases, so an answer that is semantically right but phrased outside the alias set scores as a miss. Locality is narrower still: it is a target-value non-leakage rate, and a refusal or a fluent non-answer counts as non-leaking rather than as preserved behaviour.

Natural-chain mediation remains unidentified. Section 4.3 reports why—the vehicle we specified in advance failed its own validity gate—and identifying it needs a different experiment: substituting lexically orthogonal chains under control while holding the edit fixed. We name that as future work and claim nothing from it here.

The route census rests on judge labels validated only by agreement among judges drawn from one model family, with no human-audited subsample, so a shared bias would not show up in that agreement.

Finally, the cost of the guard on unedited queries sits inside a 0.02 absolute-accuracy gate at the point estimate on full benchmark sets—.000 on 1319 GSM8K items and −.010 on 500 MATH items—while the accompanying ninety-percent intervals, $[-.022, .022]$ and $[-.061, .041]$, are wider than that gate. We therefore report point-equivalence at the tested sets and do not claim equivalence.

7 Conclusion

An edit that answers correctly on demand has not been shown to survive the reasoning the model will actually do. Evaluating the same edit at zero thinking and after a native chain exposes a gap that direct-answer validation cannot see, and the gap is edit-specific, content-dependent, and present at the largest checkpoint we tested in each of two lineages. What the failure is not is a clean erasure, and what it does admit is a chain-local causal control point.

Ethical Statement

CounterFact writes a counterfactual over a real-world value, so a reader can invert our framing: every reversion becomes the model restoring a truth we engineered away, and our suppressor becomes a tool for silencing true reasoning. At the level of frozen weights the two readings are symmetric—a corrective update and an adversarial one are the same localized parameter change, and nothing in the mechanism tells them apart. What we characterize is prior reassertion rather than truth tracking: the chain re-derives what the base network already encoded, and the unedited base shows no matching drift toward the old value (Section 3.2). The dual-use reading is real, and we treat the intervention as a research diagnostic rather than a deployable capability. Three properties bound it. It is edited-query gated: it fires only on the subject and relation whose edit is being defended, on the trigger condition defined in Section 5.1, so it is not a general

truth-suppression capability. It steers at inference time, updating no weights and applying no optimization toward concealment. And the failure it targets stays legible—by majority judge label, none of the committed reversions we censused took the associative route that leaves no verbalized trace (0 of 41)—so unguarded reversions remain auditable in the chain, though we do not claim that the absence of such a trace could serve as a monitor.

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